Sardar Patel Institute of Technology

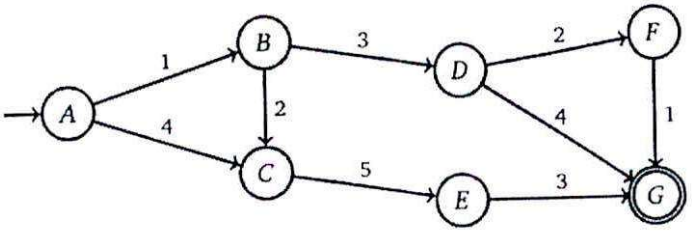
Bhavan's Campus, Munshi Nagar, Andheri (W), Mumbai : 400058, India
(Autonomous College of Affiliated to University of Mumbai)

End Semester Examination

December 2023

Max. Marks: 100
Class: Third Year
Course Code: CS307B
Name of the Course: Artificial Intelligence and Machine Learning

Duration: 3 hours
Semester: V
Branch: CE

Q. No	Question	Max Marks	CO	BL
Q1	For each of the following assertions, say whether it is true or false and support your answer with examples or counterexamples where appropriate. a. An agent that senses only partial information about the state cannot be perfectly rational. b. There exist task environments in which no pure reflex agent can behave rationally. c. There exists a task environment in which every agent is rational. d. The input to an agent program is the same as the input to the agent function. e. Every agent function is implementable by some program/machine combination. f. Suppose an agent selects its action uniformly at random from the set of possible actions. There exists a deterministic task environment in which this agent is rational. g. It is possible for a given agent to be perfectly rational in two distinct task environments. h. Every agent is rational in an unobservable environment. i. A perfectly rational poker-playing agent never loses.	20	CO 1	3
Q2	The following is a representation of a search problem, where A is the start node and G is the goal. Each edge is labelled by the cost to traverse that edge. There is also a heuristics h(n) which is defined in the table. Answer the following questions: a) Convert the above space state graph into space state tree. Hint: Let the node Names (A, B… etc) ordered in alphabet from left to right at each row or level in the tree) b) Find the path from A to G if we use: Breadth First Search (BFS), Depth First Search (DFS) c) Find the shortest path from node A to node G if we use A* search algorithm.  | n | h(n) | |---|---| | A | 5 | | B | 3 | | C | 4 | | D | 3 | | E | 3 | | F | 1 | | G | 0 |	20	CO 2	3

<table>
<tr><td>Q3a</td><td>Represent the following sentences in first-order logic, using a consistent vocabulary (which you must define):
a. Some students took French in spring 2001.
b. Every student who takes French passes it.
c. Only one student took Greek in spring 2001.
d. The best score in Greek is always higher than the best score in French.
e. Every person who buys a policy is smart.</td><td>10</td><td>CO 3</td><td>3</td></tr>
<tr><td>Q3b</td><td>There are two methods of reasoning forward and backword chaining. Discuss and differentiate between them.</td><td>10</td><td>CO 3</td><td>2</td></tr>
<tr><td>Q4</td><td>Construct a decision tree to classify the data set. Answer the following question for the resulting tree. Use information gain

<table>
<tr><th>No.</th><th>Density</th><th>Grain</th><th>Hardness</th><th>Class</th></tr>
<tr><td>1</td><td>Heavy</td><td>Small</td><td>Hard</td><td>Oak</td></tr>
<tr><td>2</td><td>Heavy</td><td>Large</td><td>Hard</td><td>Oak</td></tr>
<tr><td>3</td><td>Heavy</td><td>Small</td><td>Hard</td><td>Oak</td></tr>
<tr><td>4</td><td>Light</td><td>Large</td><td>Soft</td><td>Oak</td></tr>
<tr><td>5</td><td>Light</td><td>Large</td><td>Hard</td><td>Pine</td></tr>
<tr><td>6</td><td>Heavy</td><td>Small</td><td>Soft</td><td>Pine</td></tr>
<tr><td>7</td><td>Heavy</td><td>Large</td><td>Soft</td><td>Pine</td></tr>
<tr><td>8</td><td>Heavy</td><td>Small</td><td>Soft</td><td>Pine</td></tr>
</table>

a) What class does the tree infer for the example Density=Light, Grain=Small, Hardness=Hard?
b) What class does the tree infer for the example Density=Light, Grain=Small, Hardness=Soft?</td><td>20</td><td>CO 4</td><td>3</td></tr>
<tr><td>Q5a</td><td>Use the data below create adjacency matrix. Use single link agglomerative algorithm to cluster given dataset. Draw dendrogram.

<table>
<tr><td>Object</td><td>X</td><td>Y</td></tr>
<tr><td>A</td><td>2</td><td>2</td></tr>
<tr><td>B</td><td>3</td><td>2</td></tr>
<tr><td>C</td><td>1</td><td>1</td></tr>
<tr><td>D</td><td>3</td><td>1</td></tr>
<tr><td>E</td><td>1.5</td><td>0.5</td></tr>
</table></td><td>10</td><td>CO-4</td><td>3</td></tr>
<tr><td>Q5b</td><td>What is Density-based Clustering? What is the DBSCAN Algorithm? Write pseudo code of DBSCAN Algorithm. What are the advantages and disadvantages of the DBSCAN density-based Clustering Algorithm over other clustering techniques?

OR

What is partition based clustering? Discuss and differentiate between k-means and k-medoid clustering. Write pseudo code of k-medoids algorithm. Give a scenario and explain why use k-medoids instead of K-Means?</td><td>10</td><td>CO-4</td><td>2</td></tr>
</table>